

PROFESSIONAL SUMMARY

Software Engineer with 7+ years of experience building scalable, distributed systems across Amazon Ads and enterprise platforms. Currently an SDE 2 at Amazon, leading end-to-end development of full-stack features spanning backend services, APIs, frontend systems, and cloud infrastructure.

Proven track record of delivering complex systems including access control platforms, batch APIs, observability pipelines, and advertiser-facing products at scale. Strong expertise in system design, microservices, and cloud-native architectures (AWS/CDK).

Additionally experienced in AI/ML systems, including LLMs, RAG pipelines, and intelligent automation, with a Master's degree in Machine Learning from Carnegie Mellon.

Focused on building high-impact, reliable systems and driving engineering excellence across the stack.

EDUCATION

- **Carnegie Mellon University - School of Computer Science** Pittsburgh, PA
Master of Science in Machine Learning, School of Computer Science; GPA: 3.4/4.0 Aug. 2019 – Dec. 2020
- **Indian Institute of Technology, Kharagpur** West Bengal, India
Bachelor of Engineering in Computer Science and Engineering; GPA: 4.0 (9.72/10.0) Aug. 2013 – July. 2017

PROFESSIONAL EXPERIENCE

- **Amazon** Seattle, WA
SDE 2 – Amazon Ads Mar 2025 – Present
 - **Full-Stack Ownership of Ads Permission Delegation Platform:** Designed and delivered multiple end-to-end features spanning API models, backend services, website backend, React frontend, and CDK infrastructure across ~18 packages. Owned access level management, business justification workflows, and source validation across advertiser account management systems — leading design, implementation, and code review across the full stack.
 - **Batch APIs with Concurrent Execution & Fault Tolerance:** Architected and implemented batch ADD/DELETE APIs supporting parallel multi-threaded execution of mapping operations. Designed thread-safe processing with fine-grained concurrency control, partial success/failure isolation per mapping, and a tiered error classification system. Implemented exponential backoff with jitter for transient failures, idempotency guarantees for retries, and CloudWatch-backed status metrics for real-time batch health monitoring.
 - **ML Inference Integration & KPI Widget Platform:** Built a micro-frontend KPI visualization widget from scratch using React and Webpack Module Federation with CDN delivery, i18n support, and CDK-based deployment pipelines. Extended the widget to support dynamic metric visualization by integrating real-time ML model inference — consuming inference endpoints from a partner ML platform team — enabling live, model-driven KPI computation with sparklines and responsive layouts for Ads Console.
 - **Observability, Reliability & Security Improvements:** Built dependency service monitoring across accessor layers with CloudWatch dashboards, alarms, and structured logging, reducing mean time to detection for production incidents. Resolved Sev-2 outages via targeted validation fixes, implemented username blocklisting to address abuse vectors, and identified and mitigated dependency confusion vulnerabilities in production build pipelines.
 - **Policy-Based Access Control & Identity Infrastructure:** Refactored policy configuration layer to enable always-on policy-based access control with marketplace-aware access levels and UUID-based identity mapping. Added region validation and detailed error surfacing to improve correctness and debuggability across multi-region advertiser account systems.
 - **Operational Excellence & Technical Leadership:** Led system migrations including weblab deprecations, AMC config transitions, and feature flag removals. Authored 75+ production code reviews and reviewed 140+ peer changes, influencing API contracts, design patterns, and reliability practices across the Ads Permission Delegation ecosystem.
- **Zaytun AI (formerly LIVEr)** San Jose, CA
Senior ML Engineer Dec 2024 – Mar 2025
 - **AI-Powered Recommendation Engine:** Designed and built a vector search-based ML recommendation system leveraging LLM embeddings to match users with personalized short-form food video content and local restaurants, supporting a TikTok-style platform targeting a global Muslim audience. Owned the full ML pipeline: embedding generation, vector indexing, retrieval ranking, and relevance tuning. Migrated from Vertex AI Vector Search to Qdrant, reducing infrastructure costs and improving deployment flexibility.
 - **Scalable ML Backend for iOS Launch:** Architected and deployed backend services on Firebase and Google Cloud to support real-time ML inference delivery to mobile clients, including authentication, real-time data sync, and push notification pipelines enabling scalable content delivery and user engagement.
- **HyperWater.ai** Dallas, TX
Senior ML Engineer & Tech Lead Jan 2024 – Nov 2024

- **Generative AI & LLM Integration:** Architected and implemented an advanced ITB Preferences module leveraging LLMs and rule engines to automate decision-making processes for subcontractors. Designed the end-to-end ML pipeline from user preference ingestion to structured decision output. [Languages : Python, Java]
 - **Agent-Based Document Processing:** Built agentic systems to automate ITB document processing — applying OCR for extraction, LLM-generated decision flows for classification and routing, and integrated APIs for ITB ingestion and action-taking workflows. Owned architecture, model integration, and production deployment.
 - **LLM-Driven Workflow Optimization:** Engineered a scalable system integrating LLMs with Drools rule engine to convert unstructured user preferences into structured decision flows, improving throughput and consistency of document processing pipelines.
- **CeraMD.ai** Boston, MA
Chief Technical Officer Sep, 2023 - Dec, 2023
 - **Large-Scale AI System Architecture:** Designed and implemented a comprehensive AI-driven medical care system, integrating chat services, web crawlers, and advanced NLP models.
 - **Generative AI and RAG Implementation:** Spearheaded the development of a retriever-based system utilizing OpenAI and LLaMA models, enabling real-time, evidence-based medical insights. [Languages : Python, Node.js]
 - **Cloud Infrastructure and DevOps:** Orchestrated a robust cloud architecture on AWS, implementing microservices, serverless functions, and efficient data pipelines for scalable AI operations. [Website](#), [Web-app](#)
 - **Microsoft USA** Bellevue, WA
Software Engineer 2 May, 2021 - July, 2023
 - **Office Excel Charting Space:** Spearheaded telemetry and dashboard development, revolutionizing data visualization and analysis processes. Architected comprehensive telemetry dashboards and optimized data querying mechanisms, significantly enhancing product monitoring capabilities. [Languages : C++, C#]
 - **Azure Software Defined Networking:** Led implementation of Network Security Groups on Private Link Connections using SDN Appliance Middle box, enhancing network security and performance. [Languages : C++, C#]
 - **Qeexo** Pittsburgh, USA
ML Research Engineer Intern June 2020 – Aug 2020
 - **AutoML for Embedded Sensor Data:** Designed and prototyped real-time clustering and visualization pipelines for an AutoML platform targeting on-device and off-device execution on embedded hardware (Arduino Sensorboard). Built ML components for sensor data ingestion, unsupervised clustering, and result visualization, enabling non-expert users to train and evaluate models on resource-constrained devices.
 - **Adobe Systems** Noida, India
Senior Software Development Engineer (SDE II) June, 2017 - Aug, 2019
 - **Vector Graphics Editing (iPad):** Developed core components of a vector graphics editing application for iPad, implementing advanced tools including Pen, Direct Selection, and Snapping, enabling precise design workflows and smooth transition of desktop-grade capabilities to mobile platforms.
 - **Adobe Experience Manager (AEM) Platform:** Architected and built full-stack solutions for Interactive Communication applications, designing data integration services and document fragment-based systems to support scalable, cloud-based business process automation.
 - **Adobe Big-Data Experience Lab (BEL), Adobe Research** Bangalore, India
Research Intern May 2016 - July, 2016
 - **Channel-specific Propensity Modelling:** Developed an innovative machine learning solution for multi-channel customer behavior prediction, implementing Canonical Correlation Analysis (CCA) to enhance Adobe Analytics capabilities.

PROJECTS

- **Generative AI for Music Creation (ML Term Project, CMU):** Developed advanced deep learning models using GANs and VAE-GANs for automated music generation, demonstrating proficiency in cutting-edge AI techniques.
- **Large-Scale Text Analysis with Apache Spark (Big Data Project, CMU):** Engineered a distributed computing solution for processing and analyzing the complete works of William Shakespeare, showcasing expertise in big data technologies and scalable AI systems.
- **Entity Resolution in Large Datasets (Big Data Project, CMU):** Designed and implemented a scalable entity resolution system using Spark, applying advanced analytics techniques such as TF-IDF and cosine similarity metrics to optimize text similarity calculations across multiple data sources.
- **Machine Learning Pipeline for Song Year Prediction (Big Data Project, CMU):** Constructed an end-to-end machine learning pipeline to predict song release years based on audio features from the Million Song Dataset, optimizing linear regression models using gradient descent and grid search techniques.
- **Researcher Recommendation System (IR Term Project, IIT Kharagpur):** Developed an AI-driven search and recommendation engine for the Scientific Research Community, implementing advanced clustering algorithms to suggest potential collaborators based on co-author graph analysis.
- **Citation Analysis using Deep Neural Networks (Bachelors Thesis Project, IIT Kharagpur):** Engineered a deep learning framework to analyze and classify citation sentiment in academic papers, demonstrating proficiency in natural language processing and sentiment analysis techniques.

PUBLICATIONS

- **Towards Frequency-Based Explanation for Robust CNN:** Led research on analyzing the connection between frequency components in input data and CNN reasoning processes, providing quantitative analysis of frequency contributions to model predictions. [[Publication URL](#)]
- **Supporting Throughput Fairness in IEEE 802.11ac Dynamic Bandwidth Channel Access:** Pioneered the development of Hybrid Adaptive DBCA (HA-DBCA), an innovative mechanism for enhancing fair channel access in Dynamic Bandwidth Channel Access (DBCA) environments. [[Publication URL](#)]

KEY SKILLS

- **AI/ML:** Generative AI, LLMs, RAG systems, NLP, Computer Vision, TensorFlow, PyTorch, Scikit-learn
- **Cloud & Infrastructure:** AWS (CDK, Lambda, S3, CloudWatch), Azure, Docker, Microservices architecture
- **Programming:** Python, C++, Java, JavaScript, SQL
- **Tools & Practices:** Git, CI/CD, Agile methodologies, Data pipelines, RESTful APIs

RELEVANT COURSEWORK

- **Masters:** Machine Learning, Convex Optimization, Probabilistic Graphical Models, Deep Learning
- **Bachelors:** Information Retrieval, Natural Language Processing, Artificial Intelligence, Big Data Analytics